Spring 2018 VE300
Week 5
Lecture Notes

Cover letter = Job application letters

Solicited for advertised jobs

1. State job specifics
2. Highlight strengths in terms of reader benefits
3. Introduce resume
4. Motivate action: interview

Unsolicited Prospecting letter

1. Show interest in business
2. Introduce yourself and highlight strengths in terms of reader benefits
4. Ask about openings
5. Refer to resume
4. Motivate action: interview

Enclose	vs.	attach
Enclosure	vs.	attachment

Job application cover letters
= **persuasive & sales messages**

Effective Persuasion Techniques

1. Establish credibility
2. Make a reasonable, specific request
3. Tie facts to benefits

Sales pitch

A persuasive message designed to sell a product or service.

A **pitch** is an attempt to promote or sell.

- Get ready to *pitch* your idea to the project sponsor
- Pitch yourself to the employer based on the qualities they are looking for in an employee

Questions your readers will ask themselves:

- *Why should I?*
- *What's in it for me?*
- *What's in it for you?*
- *Who cares?*

2-minute "elevator" Pitch

Networking
In-person or in writing

5-minute pitch

In-person or phone call or screening

30+ minute pitch

Presentation or interview

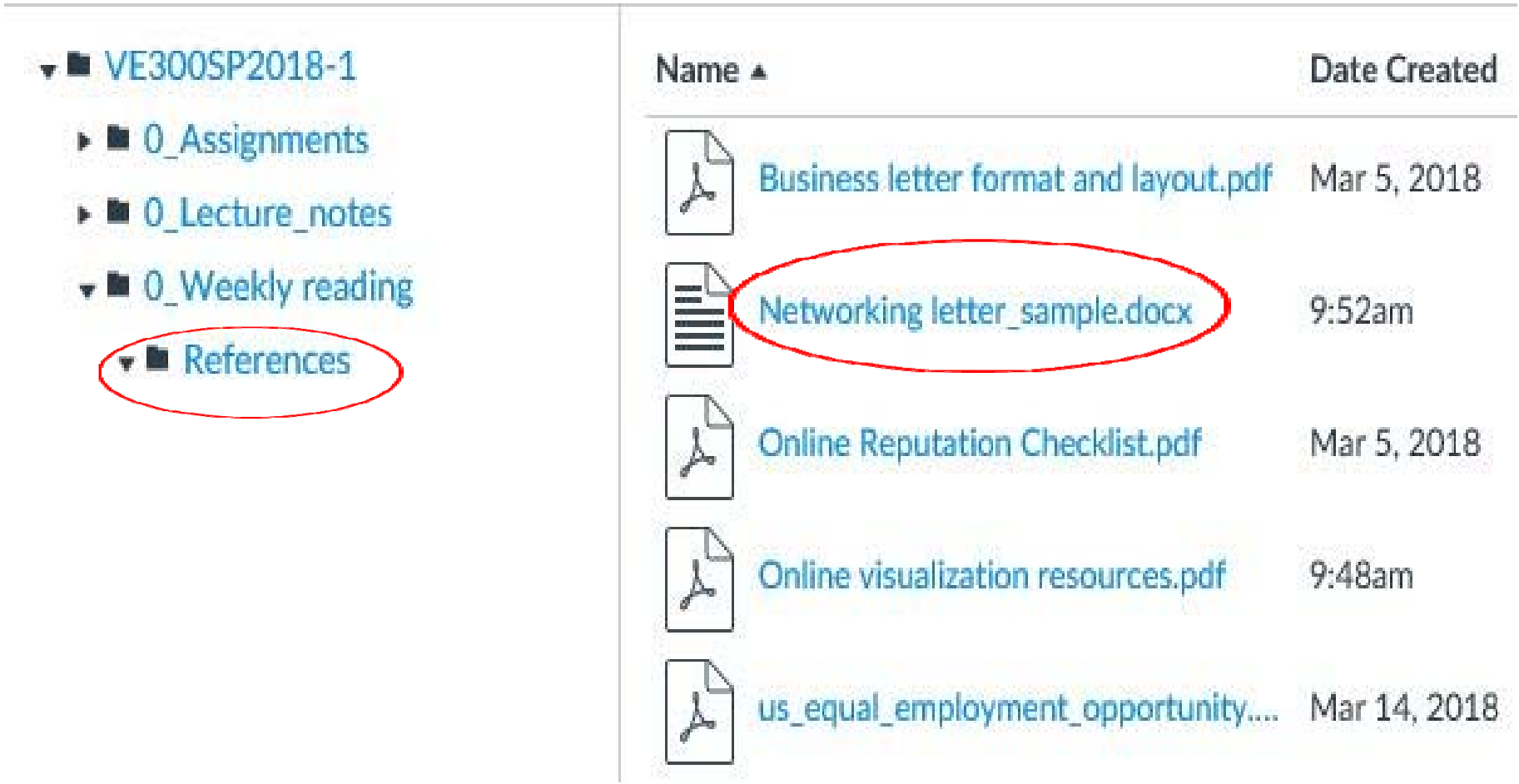
Hi John,

I enjoyed our conversation about robotic housemaids at TechCrunch Disrupt today. As I briefly mentioned, I'd love to chat with you about my startup, Deliverish, a peer-to-peer delivery marketplace where users can get anything delivered anytime anywhere.

Would you be free for coffee this Friday afternoon?

Cheers,
Erica

Networking letter sample



Being an Ethical Persuader

Persuasion becomes unethical when facts are distorted, overlooked, or manipulated with an intent to deceive.

Reading for Week 5 & 6:

CHAPTER

14

Creating and Using Graphics

Guidelines for Using Graphics *398*

Displaying Data with Graphs, Tables, and Charts *403*

Using Pictures and Drawings *410*

Using Cross-Cultural Symbols *414*

Chapter Review *417*

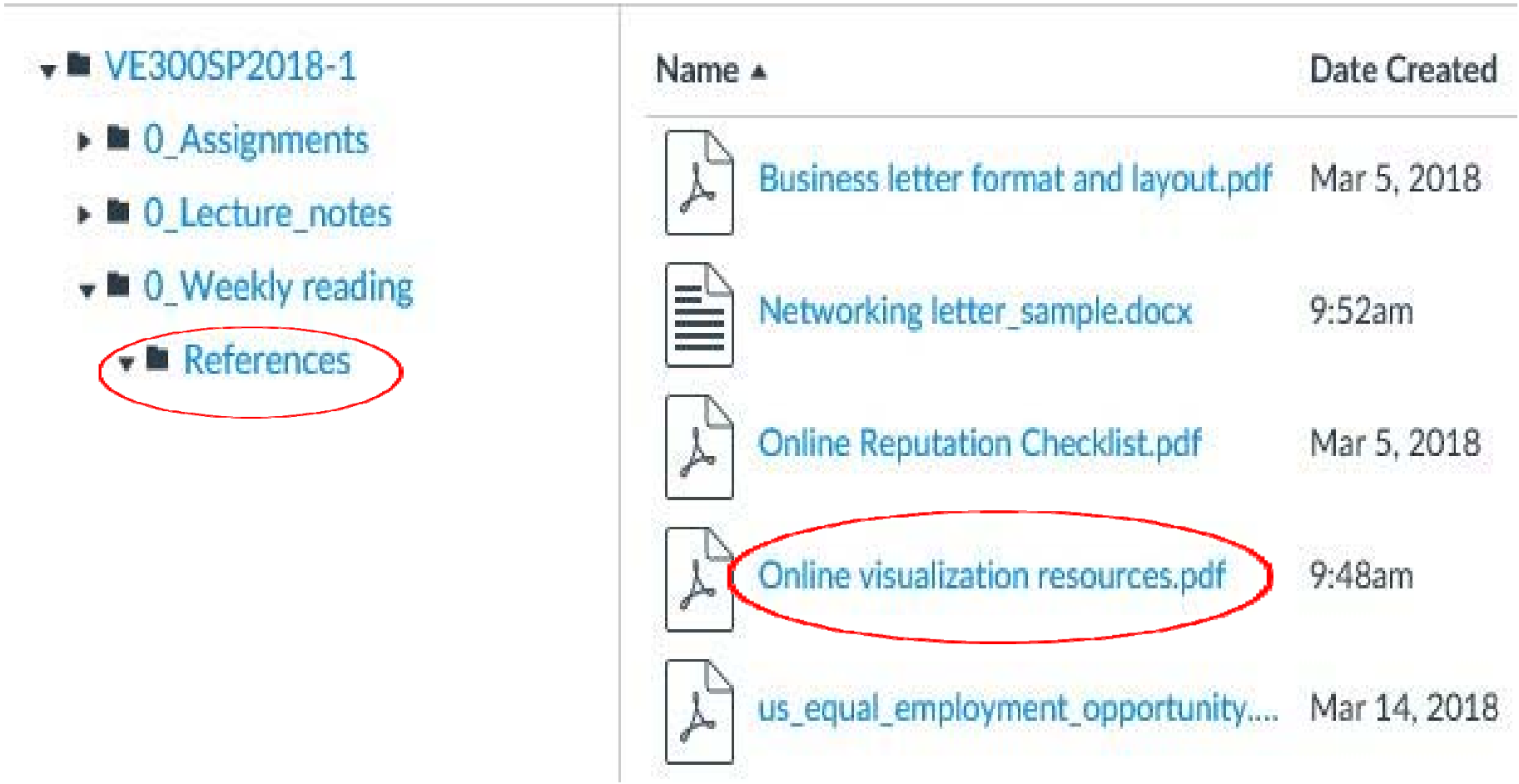
Exercises and Projects *417*

Case Study: Looking Guilty *419*

In this chapter, you will learn:

- The importance of visuals in documents and presentations.
- Four guidelines for using visuals effectively.
- How to use tables, charts, and graphs.
- Strategies for taking photographs and using them in documents and presentations.
- How drawings, icons, and clip art can be effectively used to enhance understanding.

Online visualization resources (might need VPN)



VISUAL-LITERACY.ORG

Visual Literacy: An E-Learning Tutorial on Visualization for Communication, Engineering and Business

This e-learning site focuses on a critical, but often neglected skill for business, communication, and engineering students, namely ***visual literacy***, or the ability to evaluate, apply, or create conceptual visual representations. After this tutorial, students should be able to evaluate advantages and disadvantages of visual representations, to improve their shortcomings, to use them to create and communicate knowledge, or to devise new ways of representing insights.

The didactic approach consists of rooting visualization in its application contexts, i.e. giving students the necessary critical attitude, principles, tools and feedback to develop their own high-quality visualization formats for specific problems (***problem-based learning***). The students thus learn about the commonalities of good visualization in diverse areas, but also explore the specificities of visualization in their field of specialization (through real-life case studies). They will not only learn by doing, but in doing so contribute new training material for their peers to evaluate (***peer learning***).

Project

Partners

Articles&Videos

Demo

Books & Maps

Visual literacy = Ability to create & interpret visual messages

Data
Information
Concepts
Ideas

Displaying Data with Graphs, Tables, & Charts

Choosing the appropriate graphic

The Story to Be Told	Best Graphic	How Data Are Displayed
"I want to show a trend."	Line graph	Shows how a quantity rises and falls, usually over time
"I want to compare two or more quantities."	Bar chart	Shows comparisons among different items or the same items over time
"I need to present data or facts for analysis and comparison."	Table	Displays data in an organized, easy-to-access way
"I need to show how a whole is divided into parts."	Pie chart	Shows data as a pie carved into slices
"I need to show how things, people, or steps are linked together."	Flowchart	Illustrates the connections among people, parts, or steps
"I need to show how a project will meet its goals over time."	Gantt chart	Displays a project schedule, highlighting the phases of the work

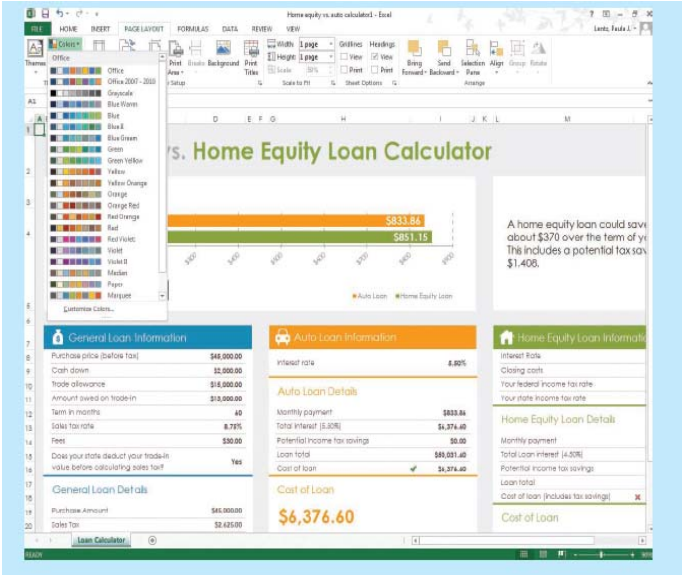
Matching graphics to objectives

Communication Needs → **Effective visual choices**

Assigned reading

Making the Most of Excel

User-friendly options for data visualization



On Canvas:
Visualization online resources

3 types of learning styles:

1. Visual
2. Auditory
3. Kinesthetic

Most people are visual learners.

Faster and better retention when information is represented and learned both visually and verbally.

Example: Too Much Text!

Hertzian Dipole Antenna Co.

These are the attractive properties of leaky-wave antennas

- Leaky wave antennas are much cheaper than phased array antennas. Phased array antennas require separate phase shifters and attenuators which must often be integrated into the antenna aperture – this is expensive!
- Leaky-wave antenna may be constructed on simple planar surfaces like circuit boards. This avoids needing cumbersome 3-dimensional structures which can cause a lot of problems.
- Leaky-wave antennas may be bent or shaped to be conformal to the surface of many types of vehicles, such as aircraft or ground vehicles. This helps improve the aerodynamic performance of the vehicles, and reduces the potential for damaging the antenna. The electrical properties of the antenna might be changed, though.
- Leak-wave antennas can be made with a higher gain than lots of types of antennas such as patches. The gain depends on the length of the leaky-wave antenna in wavelengths.
- Leaky-wave antennas are broadband due to the propagating wave nature of the current on the antenna. They may also be scanned in the elevation plane by changing the frequency within the broad band.
- Leaky-wave antennas are easy to fabricate using simple circuit-board techniques. They may be etched right onto the surface of the board and connectors may be soldered directly to the board.

Example: Nice Amount of Text

Hertzian Dipole Antenna Co.

Attractive Properties of leaky-wave antennas

- Low cost alternative to phased arrays
- Planar
- Conformal
- High gain
- Broadband (frequency scanning)
- Easy to fabricate

When to use visuals?

When to use visuals?

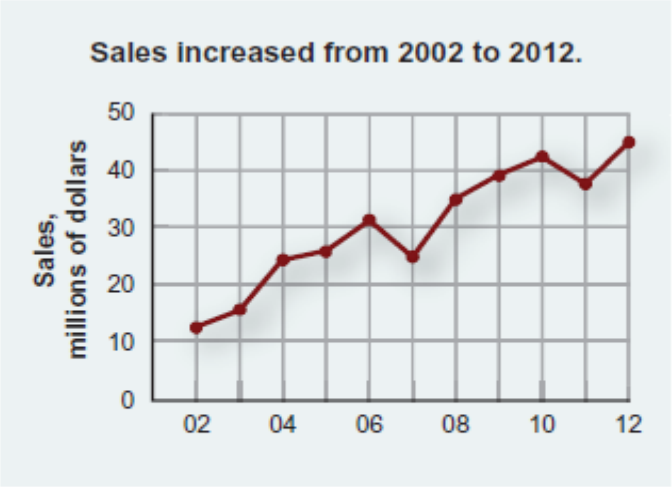
– **To clarify**

support text description of quantitative or numeric information, explanations of trends.

U.S. sales reach $44.5 million.

	Millions of dollars		
	2010	2011	2012
Northeast	10.2	10.8	11.3
South	7.6	8.5	10.4
Midwest	8.3	6.8	9.3
West	11.3	12.1	13.5
Totals	37.4	38.2	44.5

a. Tables show exact values.

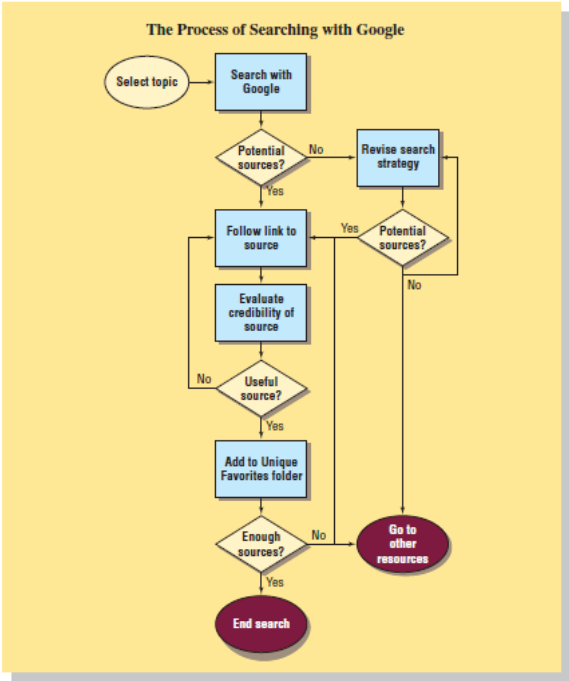


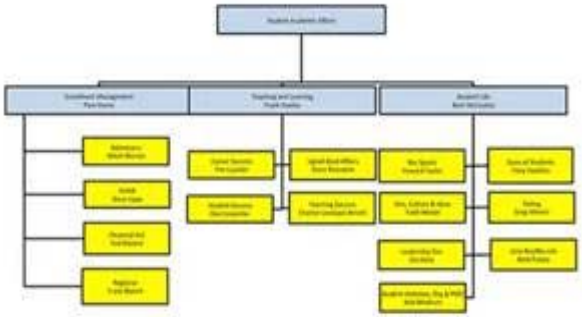
d. Line charts compare items over time or show distribution or correlation.

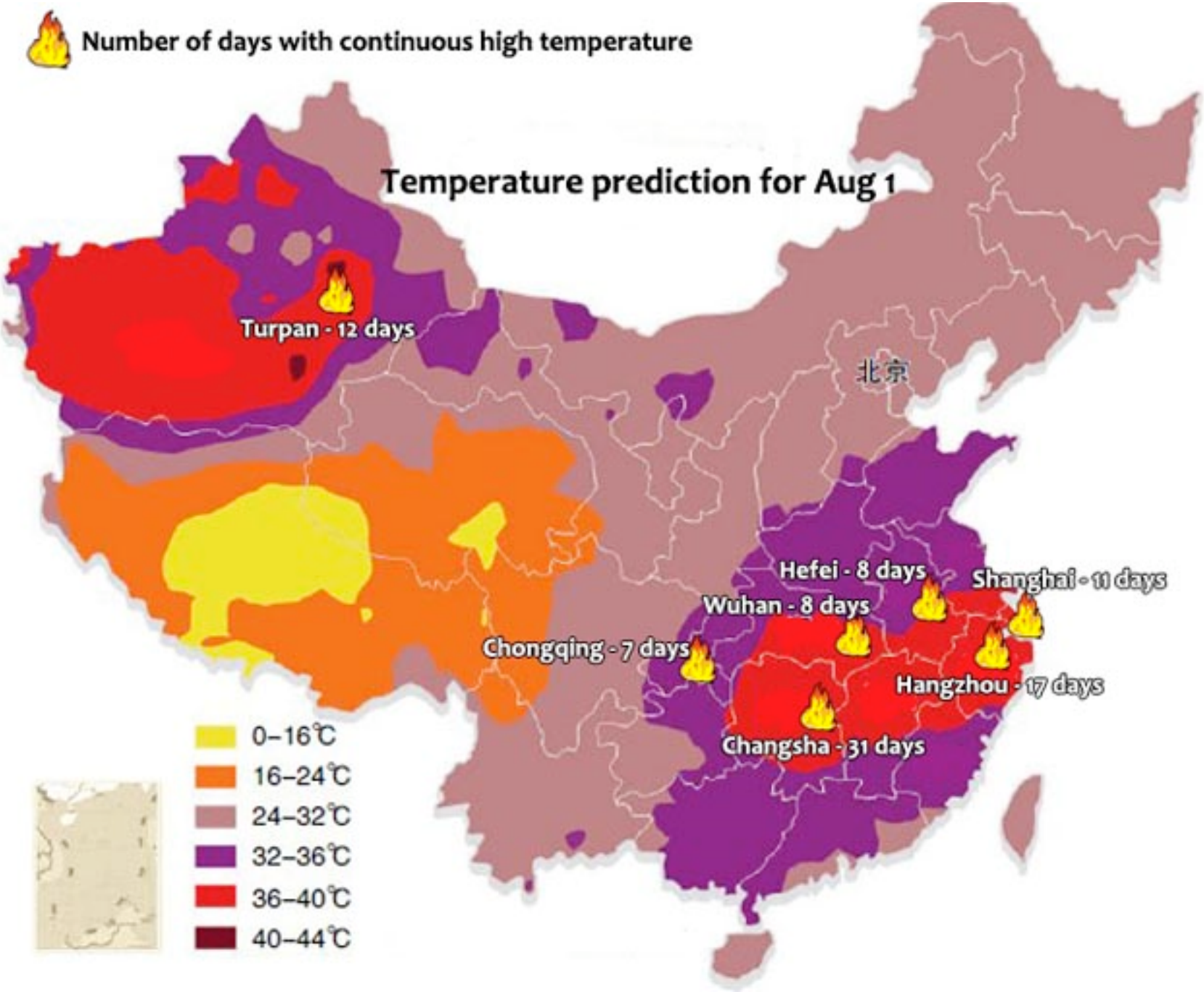
- To simplify

Divide complicated descriptions into components that can be shown with conceptual models

- Flowcharts
- Organizational charts

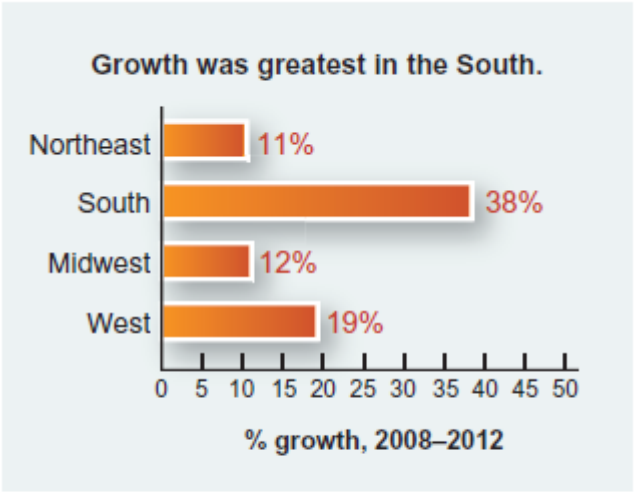




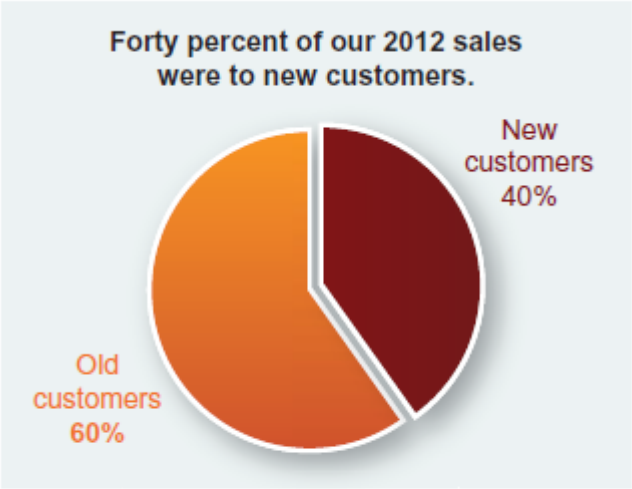

Number of days with continuous high temperature
Temperature prediction for Aug 1
Turpan - 12 days
北京
Hefei - 8 days
Shanghai - 11 days
Wuhan - 8 days
Chongqing - 7 days
Hangzhou - 17 days
Changsha - 31 days
0–16℃
16–24℃
24–32℃
32–36℃
36–40℃
40–44℃

– To emphasize

Call attention to important points by illustrating them with line, bar, and pie charts.



c. Bar charts compare items or show distribution or correlation.



b. Pie charts compare a component to the whole.

- To attract with visual symbolism

Make material seem more interesting by decorating or breaking up the text with visual aids.

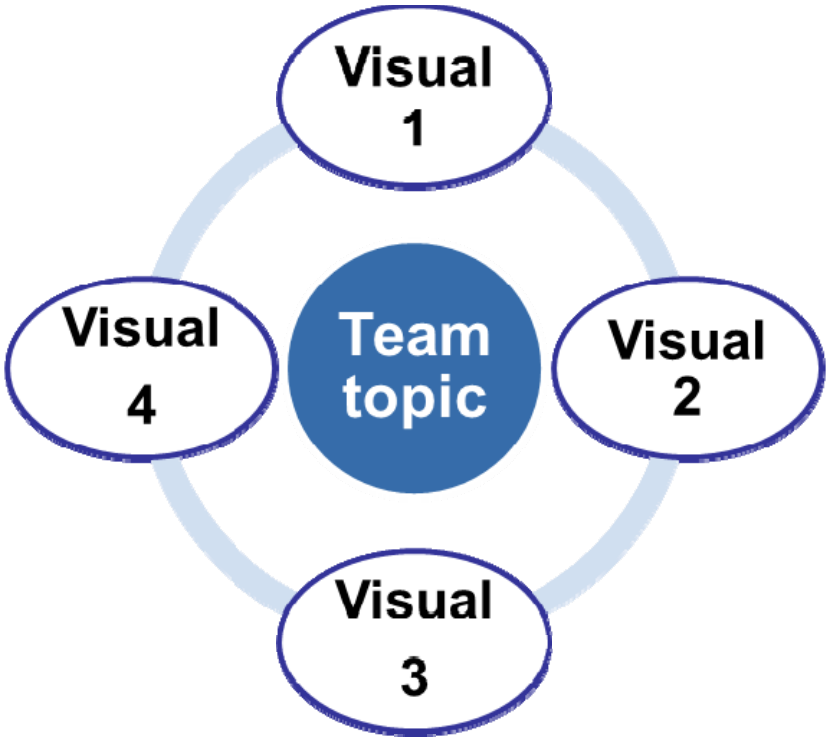
- To impress

Build credibility by using visual forms to convey authenticity and precision.

- **Video clips**
- **Cartoons**

- To unify

Show the relationships among various elements of a whole.



Planning your visuals:

- Consider aspects of your topic that visual presentation can make clear.

Visuals cannot always speak for themselves.

Make sure your audience has enough context to understand your visuals correctly.

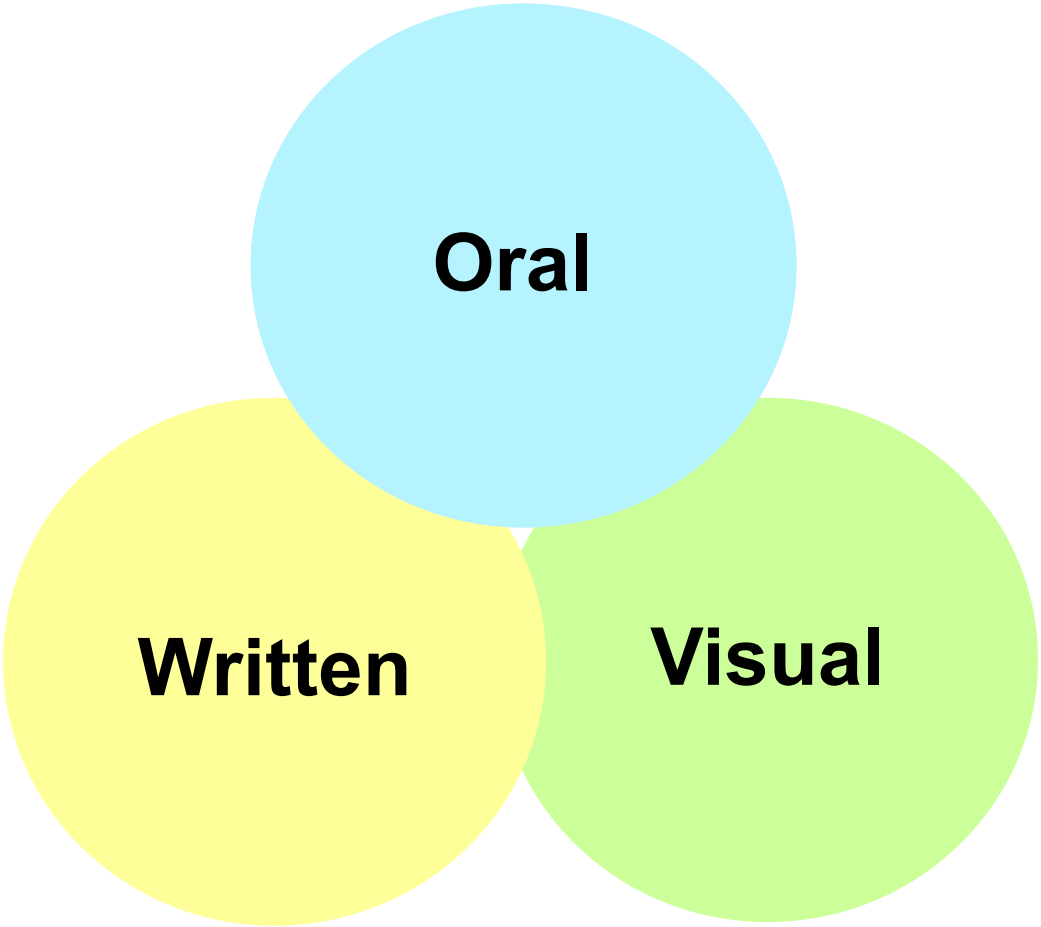
About visual aids:

- Using visuals makes the speech more interesting
- Visuals also increases your credibility.
- Visual aids should not become the presentation
- Your message is more important than the visual

Technical Communication

= convergence of multiple channels

Multi-modality



Design elements:

- Color scheme
- Font type & size
- # of main idea per slide
- # of bullet points per slide
- Position and size of visual aids

Can I use the same visuals in my report and my presentation?

No.

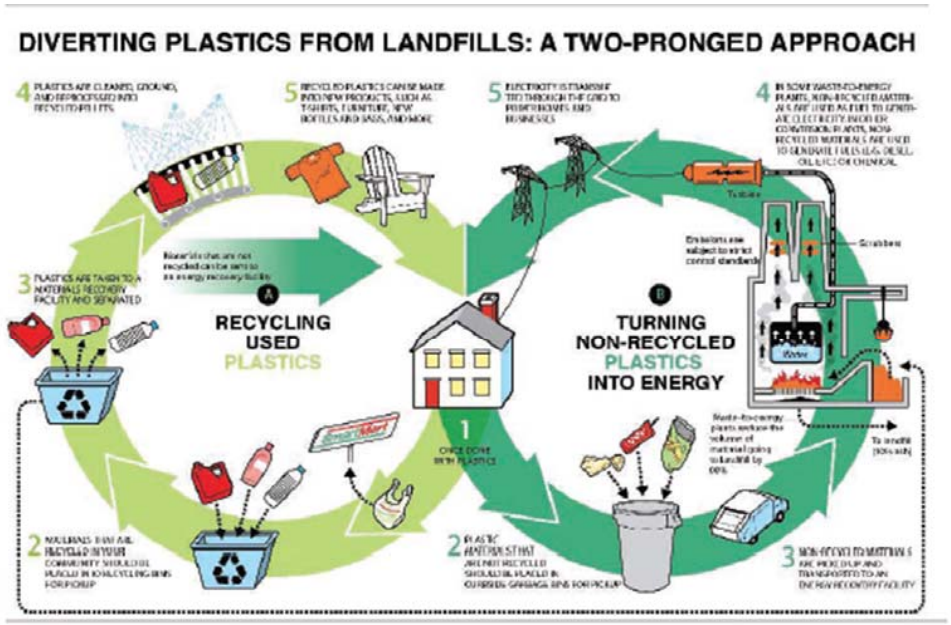
For presentations, simplify paper visuals.

The KISS principle

Keeping
It
Simple
Sweet heart

Planning visuals for presentations:

– Break a complicated point down into several visuals.



Source: From "Diverting Plastics from Landfills: A Two-Pronged Approach," *Plastics Make It Possible*, American Chemistry Council, 21 Sept. 2012, Web, 24 May 2013. http://plasticsmakeitpossible.com/2011/09/diverting-plastics-from-landfills-a-two-pronged-approach/

Planning visuals for presentations:

- Give each visual a title that makes a point.
- Don't need figure numbers

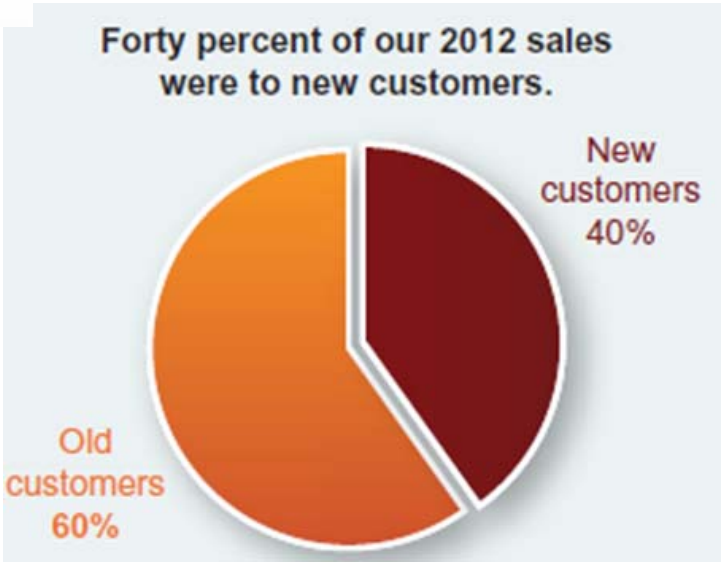
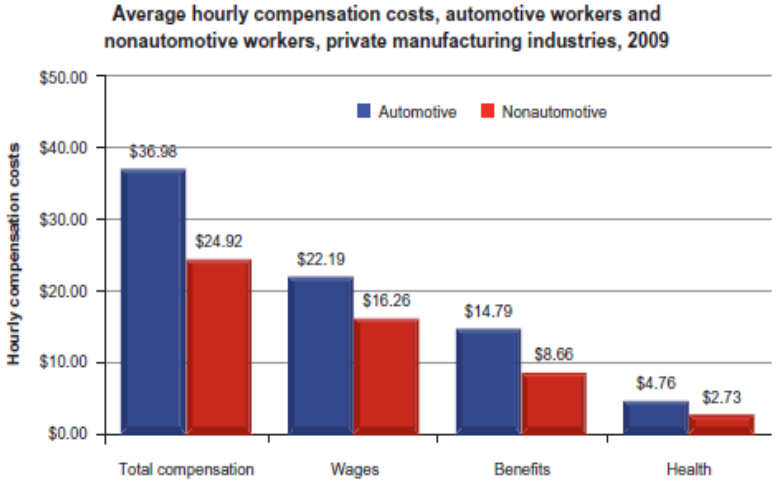
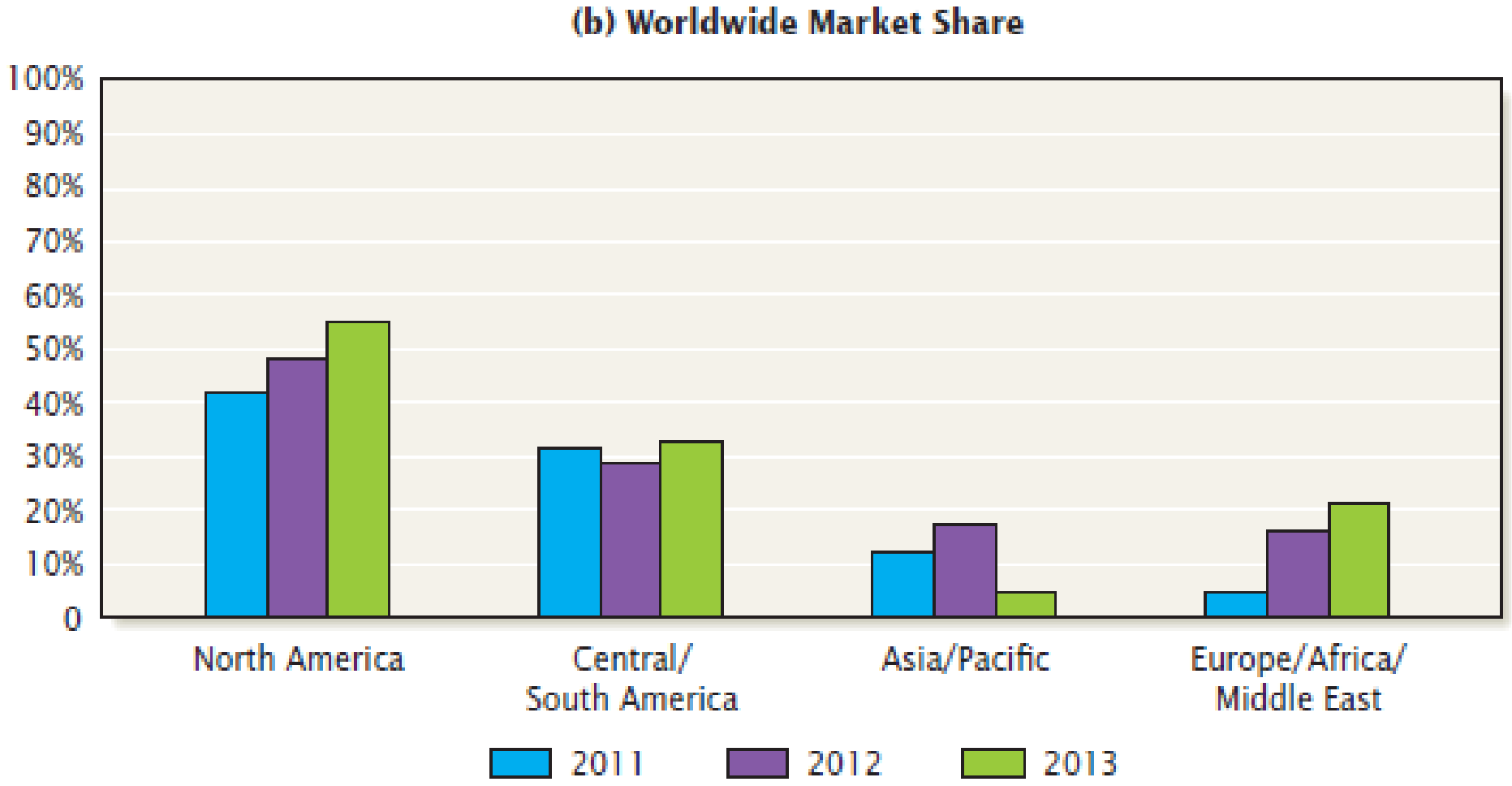


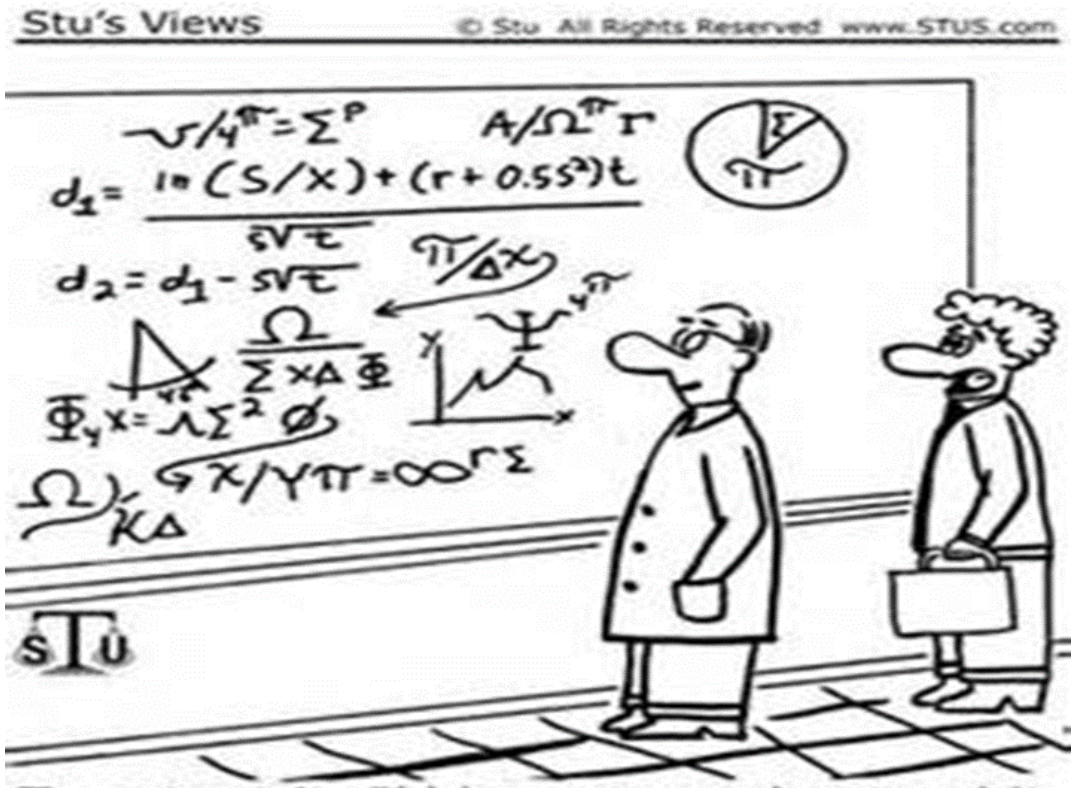
Figure 25.1 Paired Graphs Tell a Complex Story

Source: U.S. *Bureau of Labor Statistics*, April 2011.



Titles Legends





Captions

To patent it, I'd have to understand it.
You may need a different lawyer

Planning visuals for presentations:

- Limit the amount of information on a visual.
- Use simple graphs, not complex ones.

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	Millions of dollars		
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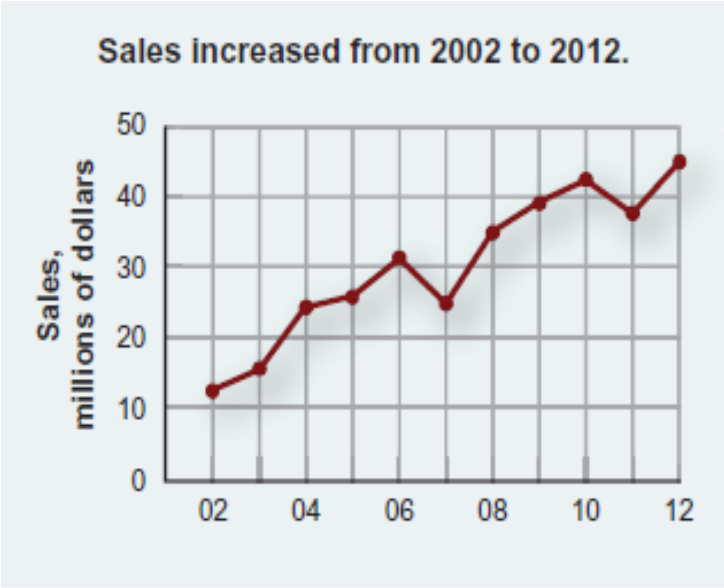
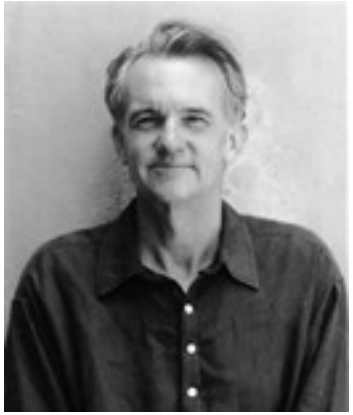




Edward Tufte (1942-)

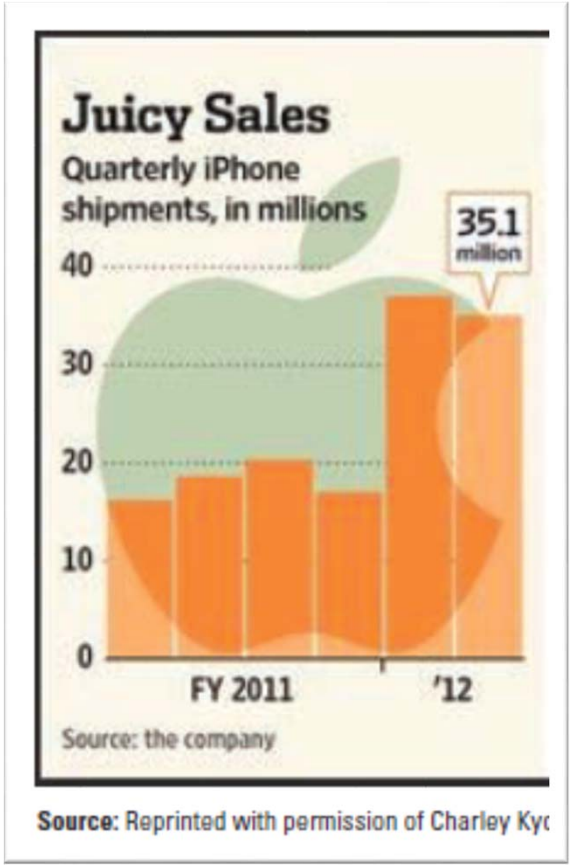
Information design
Data visualization

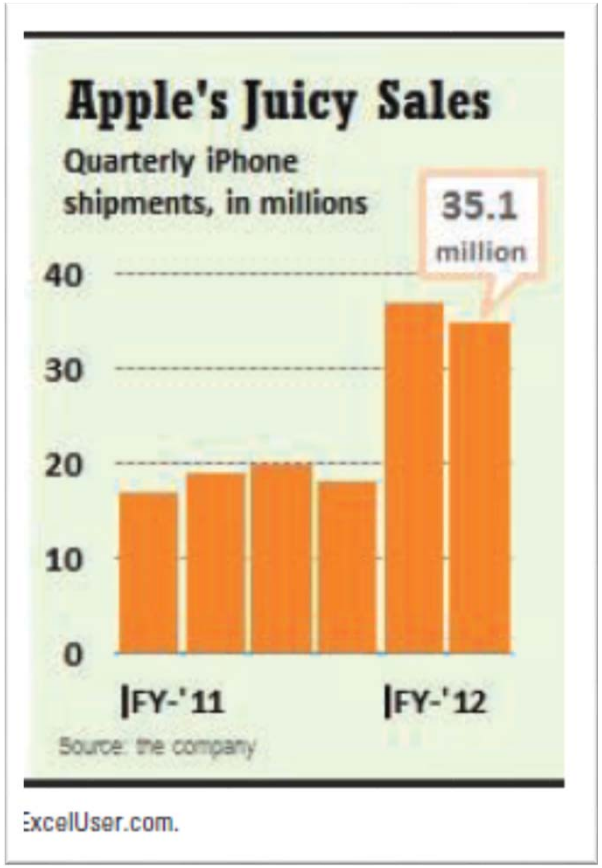
Chartjunk:

Any element in a visual that is either unnecessary or irrelevant to the reader's understanding of the data or that impede or distract from a reader's ability to understand the data.

Charts about iPhone sales

Same data, different design





Avoiding Chartjunk

What makes *chartjunk* in a visual: Those elements that "act as noise where your readers need silence."

Thank you!

www.umji.sjtu.edu.cn